

Assignment 2 Design Note

Behaviour-aware retrieve-then-rank news recommendation on MIND and EB-NeRD

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CS4.406: Information Retrieval & Extraction

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1. Problem and contribution

The task is to rank the candidate articles in a news impression by click likelihood, using article text, a user's earlier clicks, session context where available, and time-sensitive article signals. I built a reproducible pipeline for English MIND and Danish EB-NeRD data, then developed a practical two-stage system: lexical and semantic retrieval signals feed a behavioural LambdaRank re-ranker.

The report presents local temporal experiments for both datasets and a separate large-test Codabench evidence section. The shared implementation uses a normalized schema while preserving each publisher's language, metadata, and released behavioural fields. Offline development values are explicitly distinguished from hidden-test leaderboard values.

Key design decisions

- Use chronological rather than random interaction splits, so future clicks and popularity never leak into training features.
- Combine BM25, recency-weighted semantic similarity, causal popularity, freshness, category/topic affinity, and position/history signals in a LightGBM LambdaRank model.
- Keep a strict distinction between catalog retrieval experiments and in-view re-ranking, because Codabench requires a rank for every supplied candidate.
- Stream large hidden-test archives, checkpoint output, validate each rank permutation, and zip only the required root-level prediction file.

2. Data protocol and leakage boundary

Raw files are parsed into shared articles and impressions tables. An impression contains a timestamp, candidate article IDs, clicked IDs, user history, and optional session fields. Data are partitioned chronologically: old interactions train the model, the subsequent period validates choices, and the latest local period is held out. There is no random split of interaction rows.

3. Reproducible feature pipeline

Local development store	Catalog	Train impressions	Validation	Held-out test
MIND-small	65,238	126,690	30,269	73,158
EB-NeRD demo/small	11,777	42,202	3,962	3,916

Table 1. Processed temporal development stores used for fast, reproducible local experiments.

Feature construction is causal. For an event at time t , popularity counts only clicks before t ; the item clicked at t and all later clicks are excluded. Session aggregates are scanned by timestamp and use only prior impressions in the same session. MIND lacks native session and dwell-time fields, so those values are zero rather than treating a user's full history as one artificial session. For EB-NeRD, released histories are treated as upstream-causal and are never augmented from later logs.

Candidate-level feature store

Family	Signal and reason
Lexical	BM25 score over title + abstract. The query concatenates titles from the five most recent clicks.
Semantic	Recency-weighted user/article similarity. MIND uses 100-dimensional entity-derived vectors; EB-NeRD uses supplied 300-dimensional document vectors.
History preference	Category affinity, topic affinity, and history length. Recent clicks get larger weight (0.8 per step from old to new).
Article state	Causal click popularity and freshness. These model topical demand and rapid news decay.
Context	Input-list position bias and, where safely available, prior session aggregates. MIND session/dwell fields are absent; test-time unavailable fields are excluded from serving.

Table 2. Re-ranker inputs. The final submission schema is constrained to features available in released test inputs.

4. Candidate generation and ranking protocol

Stage 1 is BM25 over the complete catalog, using $k_1 = 1.5$ and $b = 0.75$. For global-catalog diagnostics it retrieves top-200 articles and excludes articles already clicked by the user. Recall is modest because news is time-sensitive and a user's next click often differs from old history.

Dataset	BM25 Recall@50 / @100 / @200	Semantic Recall@50 / @100 / @200
MIND	0.01348 / 0.02152 / 0.03219	0.00351 / 0.00621 / 0.01045
EB-NeRD	0.00959 / 0.01615 / 0.03054	0.00833 / 0.01388 / 0.02764

Table 3. Local open-catalog retrieval recall. Semantic search is a brute-force cosine baseline, not a production ANN service.

Stage 2 is an in-view LightGBM LambdaRank re-ranker: candidates are grouped by impression and binary click labels are optimized with a listwise objective. For evaluation, the code reports both normal in-view ranking and a separate retrieval-gated diagnostic. The former is the right protocol for Codabench, which supplies the candidate list and requires a complete rank permutation; it is not mislabeled as full open-catalog evaluation.

5. Re-ranker architecture and local results

The primary ranker is LightGBM LGBMRanker with the LambdaRank objective, 200 trees, 31 leaves, and a learning rate of 0.05. This choice is lightweight enough for tabular click features while allowing nonlinear interactions, such as a category match becoming more useful when an article is fresh. A histogram gradient-boosting fallback keeps the project runnable on macOS if LightGBM/OpenMP is unavailable, but all reported values below use LambdaRank.

Dataset / validation	Method	AUC	MRR	nDCG@5	nDCG@10
MIND (30,269)	BM25	0.54512	0.28294	0.25438	0.31106
MIND (30,269)	LambdaRank	0.59020	0.30192	0.27650	0.33589
EB-NeRD (3,962)	BM25	0.49933	0.31375	0.34496	0.42899
EB-NeRD (3,962)	LambdaRank	0.65124	0.40594	0.45562	0.52147

Table 4. Local chronological experiments. These are local validation metrics, not hidden-test leaderboard scores.

All four BM25-to-re-ranker improvements are statistically supported by a paired 2,000-resample bootstrap. The 95% confidence intervals for the differences are AUC [0.04125, 0.04881], MRR [0.01623, 0.02183], nDCG@5 [0.01907, 0.02508], and nDCG@10 [0.02217, 0.02750] on MIND. EB-NeRD likewise has positive reranker-versus-BM25 intervals; their exact values appear in Table 8. Every reported interval excludes zero.

Offline MINDlarge development validation

This is an offline development experiment, not a Codabench leaderboard result. MINDlarge_train trained the model and the temporally later official MINDlarge_dev split selected the configuration. Across 376,471 development impressions, the model achieved AUC 0.61820, MRR 0.34163, nDCG@5 0.32400, and nDCG@10 0.38310. The causal popularity-only reference achieved 0.53857, 0.26179, 0.24527, and 0.30787, respectively. The model metric bootstrap intervals are narrow, for example AUC [0.61734, 0.61914]. EB-NeRD local validation remains the engineering comparison reported in this note.

Interpretation

The large-dev results were used for model selection. They should not be directly compared with the smaller local MIND experiment because the data scale, split, and candidate distributions differ.

6. Why this improves the baseline

BM25 captures direct topical overlap, but it cannot model which category, entity profile, freshness level, and popularity pattern best match a particular user. The re-ranker learns these interactions from prior behavior. Semantic similarity helps when lexical terms do not overlap, while causal popularity supplies a safe demand prior for sparse histories.

Cross-dataset observations

Dataset	Lexical versus semantic finding	Effect of behavioural re-ranking
MIND	Semantic AUC is higher (0.555 vs. 0.544), but BM25 has higher nDCG@10 (0.311 vs. 0.301).	LambdaRank has the strongest local relevance results; history features improve top-rank ordering.
EB-NeRD	Semantic slightly exceeds BM25 in AUC (0.507 vs. 0.499) and nDCG@10 (0.435 vs. 0.429).	LambdaRank produces a much larger local lift, especially for fresh and preference-matched items.

Table 5. Retrieval observations from the extended local evaluation. Values are not comparable across publishers as absolute quality scores.

The lexical-semantic relationship is dataset-dependent: text overlap is relatively more useful at the top of MIND rankings, while the supplied EB-NeRD vectors offer a small standalone gain. In both datasets, the behavioural re-ranker is the main source of improvement because it can combine user preference, freshness, and popularity safely over time.

7. Large-test Codabench evidence

This section contains only submissions made with the required large hidden-test archives: MINDlarge_test and ebnerd_testset. The local small/demo and development experiments elsewhere in this report are offline diagnostics and are not leaderboard submissions.

Competition	Large test input	Submission	Status	Visible result
MIND	MINDlarge_test	mind_v2_submission.zip (ID 934568)	Finished	AUC 0.6290; MRR 0.3071; nDCG@5 0.3295; nDCG@10 0.3860
EB-NeRD	ebnerd_testset	ebnerd_personalized_submission_A2.zip (ID 934830)	Submitted	No numerical score at capture

Table 6. Large-test Codabench evidence. EB-NeRD is correctly reported as submitted, not scored.

ID # ▾	File name	Date	Status	Score	Actions
934568	mind_v2_submission.zip	2026-09-20 11:15	Finished	0.6290	 

Figure 1. MIND Codabench large-test submission: mind_v2_submission.zip (ID 934568), finished on 20 September 2026; displayed score 0.6290.

55	joepel	2026-08-17 06:07	891216	0.647	0.3129	0.3387	0.3952
56	minden0612	2026-08-22 18:46	897329	0.6469	0.3154	0.3396	0.3957
57	zitqan	2026-09-09 15:39	920008	0.6442	0.3129	0.3361	0.3924
58	aryanil	2026-08-24 05:00	898751	0.6441	0.3128	0.3361	0.3923
59	khooshi	2026-08-30 21:57	908859	0.6425	0.3138	0.3378	0.3935
60	sriharsha2608	2026-08-27 13:37	904007	0.6403	0.3033	0.3286	0.3863
61	shambhavigantla	2026-09-16 15:40	927175	0.6403	0.3107	0.3342	0.3915
62	new_lmy	2026-03-24 08:31	647667	0.6364	0.3061	0.3281	0.3848
63	dra9on	2026-09-19 18:18	933498	0.6318	0.2902	0.3093	0.3694
64	shawbidisha	2026-09-20 11:15	934568	0.629	0.3071	0.3295	0.386
65	sherley	2026-08-24 17:17	899684	0.6255	0.2992	0.322	0.3795
66	neillr	2026-08-25 09:29	900644	0.6253	0.3069	0.3296	0.3854
67	orbi8r	2026-09-19 21:10	933727	0.6225	0.3	0.3217	0.3788
68	imankc	2026-09-14 22:00	926252	0.6223	0.3016	0.3234	0.3793
69	aayush18602	2026-08-28 18:55	905985	0.6218	0.3015	0.3223	0.3781
70	parth11	2026-09-17 20:23	930394	0.6208	0.2904	0.3118	0.3683

Figure 2. MIND Codabench leaderboard row for submission ID 934568. In the official metric order, the visible values are AUC 0.6290, MRR 0.3071, nDCG@5 0.3295, and nDCG@10 0.3860.

ID # ▾	File name	Date	Status	Score	Detailed Results	Actions
934830	ebnerd_personalized_submission_A2.zip	2026-09-20 13:37	Submitted			
934109	ebnerd_personalized_submission.zip	2026-09-20 03:49	Scoring			

Figure 3. EB-NeRD Codabench large-test submission status on 20 September 2026. The A2 archive is Submitted; the earlier archive is Scoring. No EB-NeRD score was available at capture time.

8. Baseline, improvement, and statistical significance

MIND ablation: category affinity

The principal controlled MIND improvement is category affinity: the fraction of a user's recency-weighted history assigned to the candidate's category. I remove this feature, retrain on identical data, and pair full-model and ablated metrics within each impression. This isolates a behavioural preference signal rather than merely changing model capacity.

Full model minus ablated model	Delta	95% paired bootstrap CI	Zero excluded?
AUC	+0.02542	[0.02287, 0.02806]	Yes
MRR	+0.01409	[0.01192, 0.01636]	Yes
nDCG@5	+0.01572	[0.01353, 0.01798]	Yes
nDCG@10	+0.01655	[0.01455, 0.01861]	Yes

Table 7. MIND local feature ablation, 2,000 paired bootstrap resamples.

The positive confidence intervals show that category-aware matching is not a random fluctuation in this local temporal split. The internal causal temporal-popularity baseline is deliberately named as an internal reference rather than an official NRMS reproduction. The code supports strict comparison against an externally supplied starter prediction file when it is available on the identical validation split.

EB-NeRD local re-ranker improvement

Metric	BM25	LambdaRank	Delta and 95% paired CI
AUC	0.49933	0.65124	+0.15190 [0.13998, 0.16499]
MRR	0.31375	0.40594	+0.09219 [0.08079, 0.10314]
nDCG@5	0.34496	0.45562	+0.11067 [0.09829, 0.12295]
nDCG@10	0.42899	0.52147	+0.09248 [0.08246, 0.10215]

Table 8. EB-NeRD local reranker-versus-BM25 comparison, 3,962 impressions and paired bootstrap confidence intervals.

9. Extended evaluation

Dataset / local re-ranker	ILD@10	Novelty@10	Coverage@10
MIND	0.6393	15.515	0.04500
EB-NeRD	0.1775	14.262	0.06160

Table 9. Beyond-accuracy local snapshots. Raw diversity values are dataset-specific because the two embedding spaces differ.

The re-ranker improves relevance but lowers novelty relative to BM25, an expected accuracy-novelty trade-off: behavioural and popularity signals concentrate recommendations on more familiar content. MIND reports cold-versus-warm and head-versus-tail slices. EB-NeRD reports head-versus-tail slices; its local validation subset contains no cold-history impressions, so a cold slice is explicitly unavailable rather than imputed.

10. Serving and scale analysis

A local single-process benchmark measures complete request work: build the BM25 history query, retrieve around 200 catalog candidates, construct features, and score the ranker. The machine is not a production server, so these values are measured local evidence and not a cloud capacity guarantee.

Measured local quantity	MIND	EB-NeRD
BM25 postings estimate	103.2 MB	14.1 MB
Embedding index	22.5 MB (100-d)	14.1 MB (300-d)
Feature-store footprint	56.5 MB	44.5 MB
End-to-end p50 / p95 / p99	278 / 378 / 464 ms	166 / 414 / 571 ms
Single-process throughput	3.64 requests/s	5.06 requests/s
Semantic-only p99	95.2 ms	93.6 ms
1000 QPS rough estimate	275 workers; USD 137.50/h	198 workers; USD 99.00/h

Table 10. Measured local serving snapshots over 300 validation requests. The corpora and hardware profile differ, so this is a scale diagnostic, not a fair speed race.

Both end-to-end p99 values miss the illustrative 100 ms SLA, while semantic-only scoring is close to the target. This identifies Python-side lexical retrieval, per-request feature construction, and ranker invocation as the dominant bottlenecks. The cost extrapolations are intentionally conservative and should not be treated as production quotes.

What breaks first at 10x

- Dense vector search: brute-force dot products become expensive as the catalog grows. Use a measured FAISS/HNSW ANN index with recall monitoring.
- Feature computation: Pandas reconstruction of histories and popularity is too slow online. Materialize time-safe, versioned features in an online store such as Redis or Feast.
- Lexical retrieval: in-memory BM25 postings need sharding, caching, or a dedicated search service as catalog and query traffic grow.
- Ranking throughput: batch candidate matrices across requests and use concurrent workers or a model-serving endpoint instead of sequential Python calls.
- Feature freshness: popularity, embeddings, and article metadata need explicit update SLAs, or newly published articles receive stale/default signals.

11. Dataset comparison

MIND is English and supplies entity annotations that make entity-derived vectors convenient. EB-NeRD is Danish and provides document vectors, so its semantic signal comes from the supplied Word2Vec representation. Both sources share rapid news decay, but their schemas differ in language, article metadata, and session availability. The common feature API and submission validators make the experiments comparable without pretending that identical feature values have identical meaning across publishers.

12. Reproducibility and submission procedure

The repository is organized around one-command reproducibility: build the normalized store, compute retrieval features, train/re-rank, run ablations and extended evaluation, then generate validated prediction archives. Large raw data, intermediate caches, and final prediction files are excluded from Git. The final generators use bounded batches, durable checkpoints, and strict rank-list validation before archiving.

Stage	Representative command
Rebuild development store	<code>python build_pipeline.py --source-root --output data/processed</code>
Strict re-ranker reports	<code>python reranker.py --dataset mind --serving-only --strict-two-stage</code> ; repeat with <code>--dataset ebnerd</code>
Ablation and confidence intervals	<code>python ablation.py --dataset mind --serving-only</code> ; repeat with <code>--dataset ebnerd</code>
Extended evaluation	<code>python offline_evaluation.py --serving-only --strict-two-stage</code>
Large-test prediction archives	<code>python make_mind_submission.py ...</code> ; <code>python make_ebnerd_personalized_submission.py ...</code>

Table 11. Reproduction entry points. Exact data archive paths are passed as command-line arguments or environment settings.

13. Limitations and responsible interpretation

- Logged in-view metrics evaluate the supplied candidate set. The strict top-200 retrieval-gated diagnostic is reported separately rather than claiming full open-catalog performance.
- Candidate input position can encode editorial or prior-system exposure bias. It is useful predictively but should be removed or corrected when causal exposure data are available.
- MIND does not supply rich within-session dwell context. Those unavailable fields are excluded from the final test-time serving schema.
- The internal temporal-popularity baseline is a transparent reference, not a claim of reproducing the official NRMS starter implementation.
- Local temporal validation is not proof that every metric or slice will generalize identically to the hidden leaderboard data.

14. Conclusion

The project delivers a measured and reproducible news-ranking pipeline: time-safe data preparation, lexical and semantic retrieval signals, a behavioural LambdaRank re-ranker, bootstrap-tested ablations, extended ranking diagnostics, serving measurements, and validated Codabench submission generation. The large-test evidence records the finished MIND result (AUC 0.6290, MRR 0.3071, nDCG@5 0.3295, nDCG@10 0.3860) separately from offline validation, while the EB-NeRD large-test submission remains pending a score. The next engineering priority is not a larger local model alone; it is productionizing fast ANN retrieval, cached causal features, and concurrent serving while preserving the same temporal leakage boundary.

References

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